

Solutions to Problem Set 10

1. (*) Multivariate normal distribution

[4 Points]

Let $X = (X_1, \dots, X_d)^\top$ be a random vector with values in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. We say that X has the d -dimensional multivariate normal distribution with $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d \times d}$, where Σ is symmetric ($\Sigma^\top = \Sigma$) and positive definite ($u^\top \Sigma u \geq 0$ for all $u \in \mathbb{R}^d$ and $u^\top \Sigma u \Leftrightarrow u = 0$) if it is continuously distributed with the probability density function

$$f(x) = \frac{1}{(2\pi)^{\frac{d}{2}} \sqrt{\det \Sigma}} \exp \left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right), \quad x = (x_1, \dots, x_d)^\top \in \mathbb{R}^d.$$

We write $X \sim \mathcal{N}_d(\mu, \Sigma)$.

- (a) Show that f is indeed a probability density function.

Hint: Use from linear algebra that a positive definite symmetric matrix Σ is diagonalizable and has strictly positive eigenvalues. You can also use without proof the following special case of the multidimensional transformation rule:

$$\int_{\mathbb{R}^d} g(x) dx = \int_{\mathbb{R}^d} g(By) \det(B) dy$$

for an invertible positive definite matrix B , if the integrals exist.

- (b) Show that $X_1, \dots, X_d$ are independent and normally distributed with $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ if and only if $X \sim \mathcal{N}_d(\mu, \Sigma)$, where $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$.
(c) Show that for $i, j = 1, \dots, d$, one has $\mathbf{E}[X_i] = \mu_i$ and $\text{Cov}[X_i, X_j] = \Sigma_{i,j}$, using the rule (*) below.
(d) Prove (*) for $d = 2$.

Rule ():* Let $X \sim \mathcal{N}_d(\mu, \Sigma)$ with $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d \times d}$ symmetric and positive definite. Let $p \leq d$, $A \in \mathbb{R}^{p \times d}$ with full rank p , and $b \in \mathbb{R}^p$. Then $AX + b \sim \mathcal{N}_p(A\mu + b, A\Sigma A^\top)$.

- (a) Since Σ is positive definite, there exists U such that $U^\top U = I_{d \times d}$ and

$$\Sigma = U \text{diag}(\lambda_1, \dots, \lambda_d) U^\top.$$

Therefore, we can set $\Sigma^{1/2} = U \text{diag}(\lambda_1^{\frac{1}{2}}, \dots, \lambda_d^{\frac{1}{2}}) U^\top$, which fulfills $\Sigma^{1/2} \cdot \Sigma^{1/2} = I_{d \times d}$. We also see that $(\Sigma^{1/2})^\top = \Sigma^{1/2}$ and we also see that $\det(\Sigma^{1/2})^2 = \det(\Sigma) = \prod_{i=1}^d \lambda_i > 0$, so $\det(\Sigma^{1/2}) = \sqrt{\det(\Sigma)}$. Therefore:

$$\begin{aligned} & \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \exp \left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right) dx_1 \dots dx_d \\ & \stackrel{y=x-\mu}{=} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \exp \left(-\frac{1}{2} y^\top \Sigma^{-1} y \right) dy_1 \dots dy_d \\ & \stackrel{z=\Sigma^{1/2}y}{=} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \det(\Sigma^{1/2}) \exp \left(-\frac{1}{2} z^\top z \right) dy_1 \dots dy_d \\ & = \det(\Sigma)^{1/2} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \exp \left(-\frac{1}{2} \sum_{j=1}^d z_j^2 \right) dz_1 \dots dz_d \\ & = \sqrt{\det(\Sigma)} \prod_{i=1}^d \int_{-\infty}^{\infty} \exp \left(-\frac{1}{2} z_i^2 \right) dz_i = (2\pi)^{d/2} \sqrt{\det(\Sigma)}, \end{aligned}$$

which shows that f integrates to 1. Also, $f(x) \geq 0$ for every $x \in \mathbb{R}^d$.

(b) If $X_1, \dots, X_d$ are independent with $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then

$$\begin{aligned} f_X(x) &= \prod_{i=1}^d f_{X_i}(x_i) = \prod_{i=1}^d \left(\frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}\right) \right) \\ &= \frac{1}{(2\pi)^{d/2} \left(\prod_{i=1}^d \sigma_i^2\right)^{1/2}} \exp\left(-\frac{1}{2} \sum_{i=1}^d \frac{(x_i - \mu_i)^2}{\sigma_i^2}\right). \end{aligned}$$

Now with $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$, we have $\det(\Sigma) = \prod_{i=1}^d \sigma_i^2$ and

$$(x - \mu)^\top \Sigma^{-1} (x - \mu) = (x - \mu)^\top \text{diag}(\sigma_1^{-2}, \dots, \sigma_d^{-2}) (x - \mu) = \sum_{i=1}^d \frac{(x_i - \mu_i)^2}{\sigma_i^2},$$

so $f_X(x)$ is the density of a $\mathcal{N}_d(\mu, \Sigma)$ -distribution. On the other hand, if $X = (X_1, \dots, X_d)^\top$ has f_X as density, then by the same calculation above we see that

$$f_X(x) = \prod_{i=1}^d f_{X_i}(x_i),$$

so the $X_1, \dots, X_d$ are independent and $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$.

(c) We use the rule $(\star)$ with $A = e_i$, the i th unit (row) vector $e_i = (0, \dots, 0, 1, 0, \dots, 0) \in \mathbb{R}^d$, viewed as a $\mathbb{R}^{1 \times d}$ matrix of full rank. Thus

$$X_i = AX \sim \mathcal{N}(A\mu, A\Sigma A^\top).$$

We see that $A\mu = e_i^\top \mu = \mu_i$, and $A\Sigma A^\top = e_i^\top \Sigma e_i = \Sigma_{i,i}$, thus

$$X_i \sim \mathcal{N}(\mu_i, \Sigma_{i,i}).$$

We therefore see that $\mathbf{E}[X_i] = \mu_i$. For the second claim, first let $\Sigma = I_{d \times d}$, then by (b), $X_1, \dots, X_d$ are independent and $X_i \sim \mathcal{N}(\mu_i, 1)$, $i = 1, \dots, d$, thus

$$\text{Cov}[X_i, X_j] = \delta_{i,j} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

This means that $\text{Cov}[X_i, X_j] = \Sigma_{i,j}$. Now let Σ be arbitrary. We define $Y = \Sigma^{-1/2}(X - \mu)$, and by rule $(\star)$, we have

$$Y = \Sigma^{-1/2}X - \Sigma^{-1/2}\mu \sim \mathcal{N}_d(\Sigma^{-1/2}\mu - \Sigma^{-1/2}\mu, \Sigma^{-1/2}\Sigma\Sigma^{-1/2}) = \mathcal{N}_d(0, I_{d \times d}).$$

Since we already know that $\text{Cov}[Y_i, Y_j] = \delta_{i,j}$, we now have

$$\begin{aligned} \text{Cov}[X_i, X_j] &= \text{Cov}[(\Sigma^{1/2}Y)_i + \mu_i, (\Sigma^{1/2}Y)_j + \mu_j] = \text{Cov}[(\Sigma^{1/2}Y)_i, (\Sigma^{1/2}Y)_j] \\ &= \text{Cov}\left[\sum_{k=1}^d (\Sigma^{1/2})_{i,k} Y_k, (\Sigma^{1/2})_{j,\ell} Y_\ell\right] = \sum_{k,\ell=1}^d (\Sigma^{1/2})_{i,k} (\Sigma^{1/2})_{j,\ell} \underbrace{\text{Cov}[Y_k, Y_\ell]}_{=\delta_{k,\ell}} \\ &= \sum_{k=1}^d (\Sigma^{1/2})_{i,k} (\Sigma^{1/2})_{j,k} \stackrel{\Sigma^{1/2} = \text{symm.}}{=} (\Sigma^{1/2} \cdot \Sigma^{1/2})_{i,j} = \Sigma_{i,j}. \end{aligned}$$

Here we used that $\text{Cov}[c, X] = 0$ for constant c .

- (d) We set $B = \Sigma^{1/2}$, then $B^\top = B$. Then define $U = B^{-1}(X - \mu)$, then $U \sim \mathcal{N}_2(0, I_{2 \times 2})$. It is easy to see that if $X \sim \mathcal{N}_2(0, I_{2 \times 2})$, then $BU \sim \mathcal{N}_2(0, \Sigma)$ where

$$f_{BU}(x) = \frac{1}{\det(B)} f_U(B^{-1}x), \quad x \in \mathbb{R}^2.$$

Now we show the claim for $\mu = 0$ (the case $\mu \neq 0$ follows similarly).

- $A \in \mathbb{R}^{2 \times 2}$, $\det(A) \neq 0$ (since full rank), then

$$\begin{aligned} f_{AX}(x) &= f_{ABU}(x) = \frac{1}{\det(A) \det(B)} f_U(B^{-1}A^{-1}x) \\ &= \frac{1}{2\pi \det(A) \det(B)} \exp\left(-\frac{(B^{-1}A^{-1}x)^\top (B^{-1}A^{-1}x)}{2}\right) \\ &= \frac{1}{2\pi \det(A) \det(\Sigma)^{1/2}} \exp\left(-\frac{x^\top (A\Sigma A^\top)^{-1}x}{2}\right). \end{aligned}$$

- If $A \in \mathbb{R}^{1 \times 2}$, then $A = (a_1, a_2) \in \mathbb{R}^2$ so $AX = a_1X_1 + a_2X_2$. We then see that

$$\text{Var}[AX] = a_1^2 \text{Var}[X_1] + a_2^2 \text{Var}[X_2] + 2a_1a_2 \text{Cov}[X_1, X_2] = A\Sigma A^\top.$$

2. (*) Conditional expectation for continuous and discrete random variables [4 Points]

Let $(\Omega, \mathcal{F}, \mathbf{P})$ a probability space, and $X, Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ random variables.

- (a) Let X, Y be jointly continuous with joint density

$$f_{X,Y}(x, y) = e^{-y} \mathbb{1}_{\{0 \leq x \leq y\}}.$$

Calculate $\mathbf{E}[X|Y]$ and $\mathbf{E}[Y|X]$.

- (b) Let $r, p \in \mathbb{R}$ and $0 \leq r \leq p \leq 1$ and $1 - 2p + r \geq 0$. The distributions of X and Y are given by

$$\begin{aligned} \mathbf{P}[X = 1, Y = 1] &= r, & \mathbf{P}[X = 0, Y = 1] &= p - r, \\ \mathbf{P}[X = 1, Y = 0] &= p - r, & \mathbf{P}[X = 0, Y = 0] &= 1 - 2p + r. \end{aligned}$$

Determine the law of Y and the conditional expectation $\mathbf{E}[X|Y]$.

- (a) We start with $\mathbf{E}[X|Y]$. For this, we need $f_{X|Y=y}(x)$, and therefore we calculate

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx = \int_0^y e^{-y} dx = ye^{-y}$$

for $y \geq 0$. Thus

$$f_{X|Y=y}(x) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{e^{-y}}{ye^{-y}} \mathbb{1}_{\{0 \leq x \leq y\}}.$$

(we could say that X is $\mathcal{U}([0, y])$ -distributed given that $Y = y$). We see that

$$\mathbf{E}[X|Y = y] = \int_{-\infty}^{\infty} x f_{X|Y=y}(y) dx = \int_0^y \frac{x}{y} dx = \frac{y}{2}.$$

Hence $\mathbf{E}[X|Y] = \frac{Y}{2}$.

Now we move to $\mathbf{E}[Y|X]$ for this, we need $f_{Y|X=x}(y)$, and we calculate

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \int_x^{\infty} e^{-y} dy = e^{-x},$$

for $x \geq 0$. This gives

$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \frac{e^{-y}}{e^{-x}} \mathbb{1}_{\{x \leq y\}}.$$

We see that

$$\begin{aligned}\mathbf{E}[Y|X = x] &= \int_{-\infty}^{\infty} y f_{Y|X=x}(y) dy = \int_x^{\infty} y e^{-(y-x)} dy \\ &= \int_0^{\infty} (\tilde{y} + x) e^{-\tilde{y}} d\tilde{y} = x + 1.\end{aligned}$$

We conclude that $\mathbf{E}[Y|X] = X + 1$.

(b) We have that

$$\mathbf{P}[Y = 0] = p - r + 1 - 2p + r = 1 - p, \quad \mathbf{P}[Y = 1] = r + p - r = p$$

and similarly

$$\mathbf{P}[X = 0] = p - r + 1 - 2p + r = 1 - p, \quad \mathbf{P}[X = 1] = r + p - r = p$$

(i.e. X, Y are i.i.d. and $Y \sim \text{Ber}(p)$). Now

$$\mathbf{E}[X|Y = 0] = 0 \cdot \mathbf{P}[X = 0|Y = 0] + 1 \cdot \mathbf{P}[X = 1|Y = 0] = \frac{\mathbf{P}[X = 1, Y = 0]}{\mathbf{P}[Y = 0]} = \frac{p - r}{1 - p}.$$

Similarly

$$\mathbf{E}[X|Y = 1] = 0 \cdot \mathbf{P}[X = 0|Y = 1] + 1 \cdot \mathbf{P}[X = 1|Y = 1] = \frac{\mathbf{P}[X = 1, Y = 1]}{\mathbf{P}[Y = 1]} = \frac{r}{p}.$$

This means that

$$\mathbf{E}[X|Y] = \frac{p - r}{1 - p} \mathbb{1}_{\{Y=0\}} + \frac{r}{p} \mathbb{1}_{\{Y=1\}}.$$

3. σ -algebras generated by random variables and conditional expectation

The aim of this problem is to explore the concept of conditional expectation in a more general context than presented in the lecture.

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space, $Z : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ a random variable with $\mathbf{E}[|Z|] < \infty$, and $\mathcal{G} \subseteq \mathcal{F}$ a sub- σ -algebra. We say that a random variable $U : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a *version of the conditional expectation of Z given $\mathcal{G}$* if

- (I) U is $\mathcal{G}$ - $\mathcal{B}(\mathbb{R})$ -measurable, and
- (II) For every $A \in \mathcal{G}$, we have

$$\mathbf{E}[Z \mathbb{1}_A] = \mathbf{E}[U \mathbb{1}_A].$$

One writes $U = \mathbf{E}[Z|\mathcal{G}]$. When calculating $\mathbf{E}[Z|\mathcal{G}]$, one can informally view $\mathcal{G}$ as containing all events $A \in \mathcal{F}$ for which the answer “is $\omega \in A$?” is given.

- (a) i. Let $\mathcal{G}_1 = \{\emptyset, \Omega\}$. Show that $\mathbf{E}[Z|\mathcal{G}_1] = \mathbf{E}[Z]$. Interpret this result.
- ii. Now suppose that Z is $\mathcal{G}_2$ - $\mathcal{B}(\mathbb{R})$ -measurable. Show that $\mathbf{E}[Z|\mathcal{G}_2] = Z$. Interpret this result.

To make a connection to the conditional expectation discussed in the lecture, we define for $Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ the σ -algebra generated by Y as

$$\sigma(Y) = \{Y^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}.$$

- (b) Show that $\sigma(Y)$ is a σ -algebra on Ω , and that it is the smallest σ -algebra that makes Y measurable.

From now on, suppose that Z and Y are both discrete random variables.

- (c) Show that $\mathbf{E}[Z|\sigma(Y)]$ is given by $\mathbf{E}[Z|Y]$ as defined in the lecture.
- (d) Let $(\Omega, \mathcal{F}, \mathbf{P}) = (\{1, \dots, 6\}, \mathcal{P}(\{1, \dots, 6\}), \mathcal{U}(\{1, \dots, 6\}))$ and

$$\begin{aligned}Z(\omega) &= \omega, \\ Y(\omega) &= \mathbb{1}_{\{1,3,5\}}(\omega)\end{aligned}$$

- i. Write down $\sigma(Y)$ explicitly and explain its interpretation.
- ii. Calculate $\mathbf{E}[Z|Y]$.

- iii. Now let $X(\omega) = \mathbb{1}_{\{1,2\}}(\omega)$. Calculate $\mathbf{E}[Z|X]$ and $\mathbf{E}[Z|\sigma(X, Y)]$, where $\sigma(X, Y) = \sigma(\sigma(X) \cup \sigma(Y))$. Compare the values and explain the difference.

- (a) i. Note that $U = \mathbf{E}[Z]$ is $\mathcal{G}_1$ -measurable (any constant map is $\mathcal{G}_1$ -measurable), so (I) is fulfilled. Also, we have

$$\mathbf{E}[\mathbf{E}[Z] \mathbb{1}_\Omega] = \mathbf{E}[Z] = \mathbf{E}[Z \mathbb{1}_\Omega],$$

and

$$\mathbf{E}[\mathbf{E}[Z] \mathbb{1}_\emptyset] = 0 = \mathbf{E}[Z \mathbb{1}_\emptyset],$$

so (II) is fulfilled.

Interpretation: Conditioning on the σ -algebra $\{\emptyset, \Omega\}$ amounts to knowing answers to the question “is $\omega \in \emptyset$ ” (never true) and “is $\omega \in \Omega$ ” (always true). It does not contain any non-trivial information. Therefore, the best guess for Z knowing $\mathcal{G}_1$ is simply the best guess of Z (without any information), which is $\mathbf{E}[Z]$.

- ii. By assumption, (I) is fulfilled for $U = Z$. For (II), note that

$$\mathbf{E}[U \mathbb{1}_A] = \mathbf{E}[Z \mathbb{1}_A],$$

which is also fulfilled for any $A \in \mathcal{G}_2$ by assumption.

Interpretation: Z being $\mathcal{G}_2$ measurable means that $\{Z \in B\} \in \mathcal{G}_2$ for any $B \in \mathcal{B}(\mathbb{R})$. Now, conditioning on the σ -algebra $\mathcal{G}_2$ amounts to being able to answer “is $\omega \in A$ ” for any $A \in \mathcal{G}_2$. In particular, we can for every $(-\infty, a]$ determine whether $\omega \in \{Z \in (-\infty, a]\}$. By taking an appropriate intersection over countably many $\{Z < a\}$, we can therefore determine the value of Z . So the best guess of Z knowing $\mathcal{G}_2$ is Z itself (we can determine its exact value).

- (b) Note that $\Omega = Y^{-1}(\mathbb{R})$ with $\mathbb{R} \in \mathcal{B}(\mathbb{R})$, so (S1) holds.

For (S2), suppose that $A = Y^{-1}(B)$ with $B \in \mathcal{B}(\mathbb{R})$, then $A^c \stackrel{(*)}{=} Y^{-1}(B^c) \in \sigma(Y)$, since $B^c \in \mathcal{B}(\mathbb{R})$.

Finally, for (S3), let $A_n = Y^{-1}(B_n)$ with $(B_n)_{n \in \mathbb{N}} \subseteq \mathcal{B}(\mathbb{R})$. Then

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} Y^{-1}(B_n) = Y^{-1}\left(\bigcup_{n=1}^{\infty} B_n\right) \in \sigma(Y),$$

since $\bigcup_{n=1}^{\infty} B_n \in \mathcal{B}(\mathbb{R})$.

Finally, we see that if Y is $\mathcal{G}$ - $\mathcal{B}(\mathbb{R})$ -measurable for a σ -algebra $\mathcal{G}$, then $\sigma(Y) \subseteq \mathcal{G}$. This makes $\sigma(Y)$ the smallest (by inclusion) σ -algebra $\mathcal{G}$ that makes Y $\mathcal{G}$ - $\mathcal{B}(\mathbb{R})$ -measurable.

- (c) We first determine an explicit representation for $\sigma(Y)$. Let $\Omega_Y = \{y_1, y_2, \dots, y_N\}$ (if Ω_Y is countably infinite, we interpret this as $\{y_1, y_2, \dots\}$), with pairwise distinct y_j . We claim that

$$(\Delta) \quad \sigma(Y) = \left\{ \bigcup_{j \in J} \{Y = y_j\} : J \subseteq \{1, 2, \dots, N\} \right\}$$

(with $\{1, \dots, N\} = \mathbb{N}$ in the case where Ω_Y is countably infinite). Indeed, we see that

$$\bigcup_{j \in J} \{Y = y_j\} = Y^{-1}(\{y_j : j \in J\}), \quad \text{and } \{y_j : j \in J\} \in \mathcal{B}(\mathbb{R}).$$

Therefore, the right-hand side of (Δ) is in $\sigma(Y)$. On the other hand for any $B \in \mathcal{B}(\mathbb{R})$, we see that $Y^{-1}(B) = Y^{-1}(\Omega_Y \cap B)$ and we can represent $\Omega_Y \cap B = \{y_j : j \in J_B\}$ by defining $J_B = \{j \in \{1, \dots, N\} : y_j \in B\}$. This proves equality in (Δ) .

Recall now from the lecture that

$$\mathbf{E}[Z|Y] = \sum_{j \in \{1, \dots, N\}} \frac{\mathbf{E}[Z \mathbb{1}_{\{Y=y_j\}}]}{\mathbf{P}[Y=y_j]} \mathbb{1}_{\{Y=y_j\}},$$

(for simplicity we set $\frac{\mathbf{E}[X \mathbb{1}_{\{Y=y_j\}}]}{\mathbf{P}[Y=y_j]} = 0$ if $\mathbf{P}[Y=y_j] = 0$, which does not change the value).

Now note that the expression for $\mathbf{E}[Z|Y]$ is $\sigma(Y)$ -measurable, so (I) is fulfilled (more precisely, we notice that $\mathbf{E}[Z|Y]^{-1}(B) = \bigcup_{j \in K_B} \{Y = y_j\}$, where $K_B = \{j \in \{1, \dots, N\} : \mathbf{E}[Z \mathbb{1}_{\{Y=y_j\}}] / \mathbf{P}[Y=y_j] \in B\}$ for $B \in \mathcal{B}(\mathbb{R})$).

We now verify that (II) is also fulfilled. To that end, take any $A \in \sigma(Y)$. By (Δ) , we see that there is $J \subseteq \{1, \dots, N\}$ such that $A = \bigcup_{j \in J} \{Y = y_j\}$. We need to verify that

$$(\Delta\Delta) \quad \mathbf{E}[\mathbf{E}[Z|Y] \mathbb{1}_A] = \mathbf{E}[Z \mathbb{1}_A].$$

We note that

$$\mathbf{E}[Z|Y] \mathbb{1}_A = \sum_{j \in J} \frac{\mathbf{E}[Z \mathbb{1}_{\{Y=y_j\}}]}{\mathbf{P}[Y=y_j]} \mathbb{1}_{\{Y=y_j\}}.$$

This is a discrete random variable that takes the countably many values $\frac{\mathbf{E}[Z \mathbb{1}_{\{Y=y_j\}}]}{\mathbf{P}[Y=y_j]}$, $j \in J$ and each one is attained with probability $\mathbf{P}[Y=y_j]$ (and 0 with the remaining probability; in fact, some of these values may coincide, but this is fine, since we then simply add up the probabilities). We therefore have

$$\mathbf{E}[\mathbf{E}[Z|Y] \mathbb{1}_A] = \sum_{j \in J} \mathbf{P}[Y=y_j] \times \frac{\mathbf{E}[Z \mathbb{1}_{\{Y=y_j\}}]}{\mathbf{P}[Y=y_j]} = \sum_{j \in J} \mathbf{E}[Z \mathbb{1}_{\{Y=y_j\}}].$$

On the other hand, we note that

$$(Z \mathbb{1}_A)(\omega) = \begin{cases} Z(\omega), & \text{if } Y(\omega) = y_j, j \in J \\ 0, & \text{else} \end{cases}$$

We can now see that $\mathbf{E}[Z \mathbb{1}_A] = \sum_{j \in J} \mathbf{E}[Z \mathbb{1}_{\{Y=y_j\}}]$.

- (d) i. $\sigma(Y) = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \Omega\}$. The information in $\sigma(Y)$ is exactly the information contained in knowing the answer to the questions “is ω even” and “is ω odd”.
- ii. We have

$$\frac{\mathbf{E}[Z \mathbb{1}_{\{Y=0\}}]}{\mathbf{P}[Y=0]} = \frac{\mathbf{E}[Z \mathbb{1}_{\{2,4,6\}}]}{\mathbf{P}[\{2,4,6\}]} = \frac{\frac{1}{6} \times 0 + \frac{1}{6} \times 2 + \frac{1}{6} \times 0 + \frac{1}{6} \times 4 + \frac{1}{6} \times 5 + \frac{1}{6} \times 6}{1/2} = 4.$$

Similarly,

$$\frac{\mathbf{E}[Z \mathbb{1}_{\{Y=1\}}]}{\mathbf{P}[Y=1]} = 3.$$

We see that

$$\mathbf{E}[Z|Y] = 4 \mathbb{1}_{\{Y=0\}} + 3 \mathbb{1}_{\{Y=1\}}.$$

- iii. Similarly as before, we see that

$$\mathbf{E}[Z|X=0] = \frac{9}{2}, \quad \mathbf{E}[Z|X=1] = \frac{3}{2},$$

so

$$\mathbf{E}[Z|X] = \frac{9}{2} \mathbb{1}_{\{X=0\}} + \frac{3}{2} \mathbb{1}_{\{X=1\}}.$$

Now consider $\sigma(X, Y) = \sigma(\sigma(X) \cup \sigma(Y))$. First note

$$\sigma(X) = \{\emptyset, \{1, 2\}, \{3, 4, 5, 6\}, \Omega\},$$

thus

$$\sigma(X) \cup \sigma(Y) = \{\emptyset, \{1, 2\}, \{1, 3, 5\}, \{2, 4, 6\}, \{3, 4, 5, 6\}, \Omega\}.$$

Note in passing that this is *not* yet a σ -algebra. We have

$$\begin{aligned}\sigma(X, Y) &= \sigma(\sigma(X) \cup \sigma(Y)) \\ &= \{\emptyset, \{1\}, \{2\}, \{3, 5\}, \{4, 6\}, \{1, 2\}, \{1, 3, 5\}, \{1, 4, 6\}, \{2, 3, 5\}, \{2, 4, 6\}, \\ &\quad \{3, 4, 5, 6\}, \{1, 2, 3, 5\}, \{1, 2, 4, 6\}, \{2, 3, 4, 5, 6\}, \{1, 3, 4, 5, 6\}, \Omega\}.\end{aligned}$$

In principle we could now simply “guess” a formula for $\mathbf{E}[Z|\sigma(X, Y)]$ and verify that it satisfies (I), (II); however, it is more instructive to consider the discrete random variable $(X, Y) : (\Omega, \mathcal{F}, \mathbf{P}) \rightarrow (\{0, 1\}^2, \mathcal{P}(\{0, 1\}^2))$. We note that

$$\sigma((X, Y)) = (X, Y)^{-1}(\{(0, 0), (0, 1), (1, 0), (1, 1)\}) = \sigma(X, Y).$$

Now we use the result from (c):

$$\begin{aligned}\mathbf{E}[Z|\sigma(X, Y)] &= \mathbf{E}[Z|(X, Y)] \\ &= \mathbf{E}[Z|(X, Y) = (0, 0)]\mathbb{1}_{\{(X, Y)=(0, 0)\}} + \mathbf{E}[Z|(X, Y) = (1, 0)]\mathbb{1}_{\{(X, Y)=(1, 0)\}} \\ &\quad + \mathbf{E}[Z|(X, Y) = (0, 1)]\mathbb{1}_{\{(X, Y)=(0, 1)\}} + \mathbf{E}[Z|(X, Y) = (1, 1)]\mathbb{1}_{\{(X, Y)=(1, 1)\}} \\ &= \mathbf{E}[Z|\{4, 6\}]\mathbb{1}_{\{4, 6\}} + \mathbf{E}[Z|\{3, 5\}]\mathbb{1}_{\{3, 5\}} + \mathbf{E}[Z|\{1\}]\mathbb{1}_{\{1\}} + \mathbf{E}[Z|\{2\}]\mathbb{1}_{\{2\}} \\ &= 5\mathbb{1}_{\{4, 6\}} + 4\mathbb{1}_{\{3, 5\}} + \mathbb{1}_{\{1\}} + 2\mathbb{1}_{\{2\}}.\end{aligned}$$

Interpretation: The underlying random experiment is “rolling a fair die once”, and yields $\omega \in \Omega$. We are interested in $Z(\omega) = \omega$, i.e. the outcome itself. Suppose now however that we do not observe Z directly. Instead, we are only given access to “partial information”:

- The values of $Y = \mathbb{1}_{\{1, 3, 5\}}$ are 1 and 0, and this stands for knowing the answer to the question “is $\omega \in \{1, 3, 5\}$ ” (or put differently “is ω an odd number?”). Correspondingly $\sigma(Y)$ contains exactly the information obtained by knowing the answer to this questions.
- Similarly, the values of $X = \mathbb{1}_{\{1, 2\}}$ stand for knowing the answer to the question “is $\omega \in \{1, 2\}$?” (or put differently “is ω smaller or equal to 2?”).
- The σ -algebra $\sigma(X, Y)$ is defined as the σ -algebra generated by “the information contained in both X and Y ”? Correspondingly, it should contain all events that can be determined knowing *both* the information whether ω is smaller or equal to 2 and whether ω is an odd number. Indeed, we could rewrite $\sigma(X, Y) = \sigma((X, Y))$, with the right hand side being the σ -algebra generated by the vector (X, Y) . Notice that this vector can take the 4 values $(0, 0), (0, 1), (1, 0), (1, 1)$ which stand for an answer no/no, no/yes, ... to the two questions “is ω smaller or equal to 2?” and “is ω an odd number?”. (Note also that $\sigma(X, Y)$ also contains further events such as $\{1, 2, 4, 6\}$, which stands for “ ω is even or smaller or equal to 2”: such an event does not give further information, but it *can* be determined by knowing the values of X, Y).

With this interpretation at hand, one can understand (hopefully) a bit clearer the interpretation of $\mathbf{E}[Z|\mathcal{G}]$:

- In the case where we know $\sigma(X, Y)$, we know more information, namely the answers to both questions “is ω smaller or equal to 2?” and “is ω an odd number?”. Consequently, the conditional expectation of $\mathbf{E}[Z|\sigma(X, Y)]$ is $\sigma(X, Y)$ -measurable, and gives the best guess of Z given this information. We also note that this random variable is not $\sigma(X)$ -measurable, which is plausible since we cannot determine its value only using the information contained in $\sigma(X)$.
- If we only have access to $\sigma(X) \subsetneq \sigma(X, Y)$, we have less information (our σ -algebra is smaller), namely we only know the answer to “is ω an even number?”. The $\sigma(X)$ -measurable $\mathbf{E}[Z|\sigma(X)]$ is our best guess knowing this (even less precise) information. Note also that $\mathbf{E}[Z|X]$ is in fact $\sigma(X, Y)$ -measurable, which means that we can determine its value knowing $\sigma(X, Y)$.

Finally, using (a) ii. we also see the interesting “tower property”:

$$\mathbf{E}[\mathbf{E}[Z|\sigma(X)]|\sigma(X, Y)] = \mathbf{E}[Z|\sigma(X)].$$

One can also show

$$\mathbf{E}[\mathbf{E}[Z|\sigma(X, Y)]|\sigma(X)] = \mathbf{E}[Z|\sigma(X)].$$

4. Random powers of random variables

Let X and Y be two independent random variables following Poisson distributions with parameters $\alpha (> 0)$ and $\beta (> 0)$, respectively. Consider $Z = X^Y$.

- (a) Calculate $\mathbf{E}[Z]$.

Hint: It is useful to condition on the values of one of the random variables.

- (b) Note that $\mathbf{E}[Z]$ can be viewed a function of α and β . Does it increase as α and β increase?

- (a) We use the formula

$$\mathbf{E}[Z] = \mathbf{E}[X^Y] = \mathbf{E}[\mathbf{E}[X^Y|X]],$$

see Theorem 9.7 (i). Now

$$\begin{aligned} \mathbf{E}[X^Y|X = x] &= \mathbf{E}[x^Y] = \sum_{k=0}^{\infty} \frac{x^k \beta^k}{k!} e^{-\beta} \\ &= e^{-\beta} \sum_{k=0}^{\infty} \frac{(x\beta)^k}{k!} \\ &= e^{\beta(x-1)} \end{aligned}$$

(this is the probability generating function of Y at x , namely $G_Y(x)$). Therefore, we have

$$\begin{aligned} \mathbf{E}[Z] &= e^{\beta(X-1)} \\ &= e^{-\beta} \sum_{\ell=0}^{\infty} e^{\beta\ell} \frac{\alpha^\ell}{\ell!} e^{-\alpha} \\ &= \exp(\alpha(e^\beta - 1) - \beta), \end{aligned}$$

having used the result from Problem 3 (a).

- (b) Consider

$$\phi(\alpha, \beta) = \log \mathbf{E}[Z],$$

and notice that $\mathbf{E}[Z]$ increases if and only if ϕ increases. Now

$$\frac{\partial \phi}{\partial \alpha} = e^\beta - 1,$$

which is positive for $\beta > 0$. Moreover, for $\beta = 0$, we have $\mathbf{E}[X^0] = 1$, so $\mathbf{E}[Z]$ always increases with α .

Now consider

$$\frac{\partial \phi}{\partial \beta} = \alpha e^\beta - 1,$$

this partial derivative is negative whenever $\alpha < e^{-\beta}$. Now $0 < e^{-\beta} \leq 1$, and it is possible for $\frac{\partial \phi}{\partial \beta}$ is negative when $\alpha < 1$.